

Neonatal Respiratory distress

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Outline

- Approach to neonatal respiratory distress
- Common cause of neonatal respiratory distress(TTN,RDS,MAS)
- Less common cause of neonatal respiratory distress (congenital pneumonia, PPHN, Pulmonary Hemorrhage, Pulmonary Air Leak, congenital diaphragmatic hernia)

Objectives

- To understand and diagnose the most common causes of respiratory distress in the newborn infant.
- Recognize clinical symptoms and radiographic patterns that reflect TTN, neonatal pneumonia, RDS, and MAS.
- Identify the short- and long-term complications associated with common neonatal respiratory disorders
- Understand management strategies for TTN, pneumonia, RDS, and MAS.
- Over view of up-to-date recommendations for the prevention of neonatal pneumonia, RDS, and MAS.

Approach...

- ❖ One of the following is clinical criteria to say a newborn is in RD:-
 - tachypnea >60 breath per minute
 - severe chest indrawing
 - central cyanosis
 - grunting
 - retraction(subcostal, intercostal, suprasternal)

Approach

❖ DIAGNOSIS

- is based primarily upon the **history and radiographic** findings, **as the physical findings are often similar** among the different pulmonary conditions.
- The **evolving** course also helps define the specific disorder

Approach...

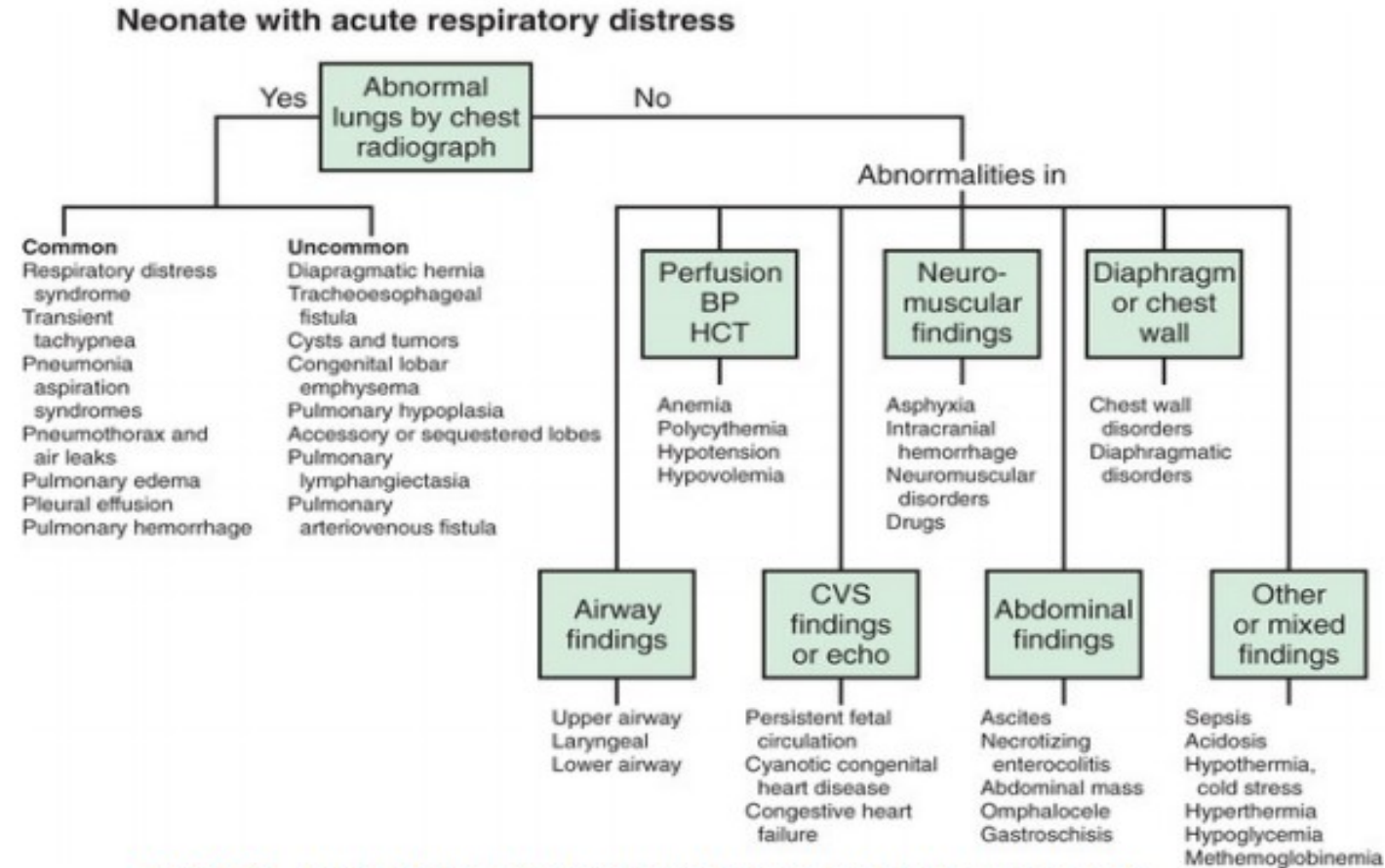


FIG. 122.1 Neonate with acute respiratory distress. BP, Blood pressure; CVS, cardiovascular system; HCT, hematocrit. (From Battista MA, Carlo WA: Differential diagnosis of acute respiratory distress in the neonate. In Frantz ID, editor: *Tufts University of School of Medicine and Floating Hospital for Children reports on neonatal respiratory diseases*, vol 2, issue 3, Newtown, PA, 1992, Associates in Medical Marketing Co.)

Approach...

- **History** —gestational age, method of delivery, risk of infection, and associated complications;
- ✓ **Transient tachypnea of the newborn (TTN)**
 - ✓ in the *late preterm infant after caesarean delivery without labor*,
 - ✓ because of the failure to initiate the normal physiologic mechanisms that contribute to lung fluid clearance.
- ✓ **Respiratory distress syndrome (RDS) –**
 - ✓ in *Preterm infants* and the risk rises as gestational age decreases
 - ✓ in infants of *diabetic mothers* compared with infants of *non-diabetic mothers* *at similar gestational age.*

Approach...

- ✓ **Persistent pulmonary hypertension (PPHN)** - Infants born through *MSAF* and those who *have perinatal depression*.
 - in infants with a history of bacterial infection, poor intrauterine growth, and NFHR patterns, suggesting poor placental function and chronic fetal hypoxemia.
 - *diagnosis* should be confirmed by echocardiography to distinguish PPHN from structural cyanotic heart disease in term infants with severe cyanosis,

Approach...

- **Chest radiography** —useful in differentiating among the disorders of neonatal respiratory distress:
- ✓ **TTN** - usually exhibits characteristic
 - Bilateral perihilar linear streaking secondary to engorged lymphatic or blood vessels.
 - Patchy infiltrates that clear within 24 to 48 hours may also reflect the fluid retention of TTN, but make initial differentiation from pneumonia problematic.
 - *Lung ultrasound* has been proposed as an imaging technique for *reliable* early diagnosis and differentiation of TTN
- ✓ **RDS**- atelectasis results in the classical radiographic findings of a diffuse, reticulogranular, ground glass appearance with air bronchograms, and low lung volume.
- ✓ **PPHN** - depends upon the presence of *associated lung disease*.
 - ✓ In infants without lung disease, the lung fields may appear clear with **decreased** pulmonary vascularity. The heart size may be normal or increased.

Approach...

TABLE 2. Developmental Stages of Lung Development and Respiratory Disease Pathogenesis

DEVELOPMENTAL STAGE	EMBRYONIC	PSEUDOGLANDULAR	CANALICULAR	TERMINAL SAC	ALVEOLAR
Gestation	0–6 weeks	7–16 weeks	17–24 weeks	25–36 weeks	>37 weeks
Structural morphogenesis	Trachea, bronchi	Bronchioles, terminal bronchioles, lung circulation	Respiratory bronchioles, primitive alveoli	Alveolar ducts, thin-walled alveolar sacs, increasing functional type 2 cells ^a	Definitive alveoli and mature type 2 cells ^a
Disease manifestation	Tracheoesophageal fistula, pulmonary sequestration	Bronchogenic cyst, congenital diaphragmatic hernia, congenital cystic adenomatoid malformation	Pulmonary hypoplasia, RDS, BPD, alveolar capillary dysplasia	RDS, BPD	TTN, MAS, neonatal pneumonia, PPHN

BPD=bronchopulmonary dysplasia; MAS=meconium aspiration syndrome; PPHN=persistent pulmonary hypertension of the newborn; RDS=respiratory distress syndrome; TTN=transient tachypnea of the newborn;

^aType 2 pneumocytes are surfactant-producing cells

Approach....

Table 3. Distinguishing Features of TTN, RDS, and MAS

<i>Cause</i>	<i>Etiology</i>	<i>Timing of delivery</i>	<i>Risk factors</i>	<i>Clinical features</i>	<i>Chest radiography findings</i>	<i>Treatment</i>	<i>Prevention</i>
TTN	Persistent lung fluid	Any	Cesarean delivery ⁴ Macrosomia Male sex Maternal asthma ² Maternal diabetes ³	Tachypnea Often no hypoxia or cyanosis	Parenchymal infiltrates ⁵ “Wet silhouette” around the heart ⁵ Intralobar fluid accumulation ⁵	Supportive, oxygen if hypoxic	Prenatal corticosteroids before cesarean delivery if 37 to 39 weeks’ estimated gestation (not accepted U.S. practice) ¹⁹
RDS	Surfactant deficiency Lung under-development	Preterm	Male sex ⁷ Maternal diabetes ⁸ Preterm delivery ⁶	Tachypnea Hypoxia Cyanosis	Homogenous infiltrates ⁵ Air bronchograms ⁵ Decreased lung volumes	Resuscitation, oxygen, ventilation, surfactant	Prenatal corticosteroids if risk of preterm delivery (24 to 34 weeks’ estimated gestation) ²⁰ (accepted U.S. practice)
MAS	Lung irritation and obstruction	Term or post-term	Meconium-stained amniotic fluid Post-term delivery	Tachypnea Hypoxia	Patchy atelectasis ⁵ Consolidation ⁵	Resuscitation, oxygen, ventilation, surfactant	Do not impede delivery for suctioning ²³ ; amnioinfusion of no benefit ²⁷

TTN = transient tachypnea of the newborn; RDS = respiratory distress syndrome; MAS = meconium aspiration syndrome.

Information from references 2 through 8, 19, 20, 23, and 27.

Approach...

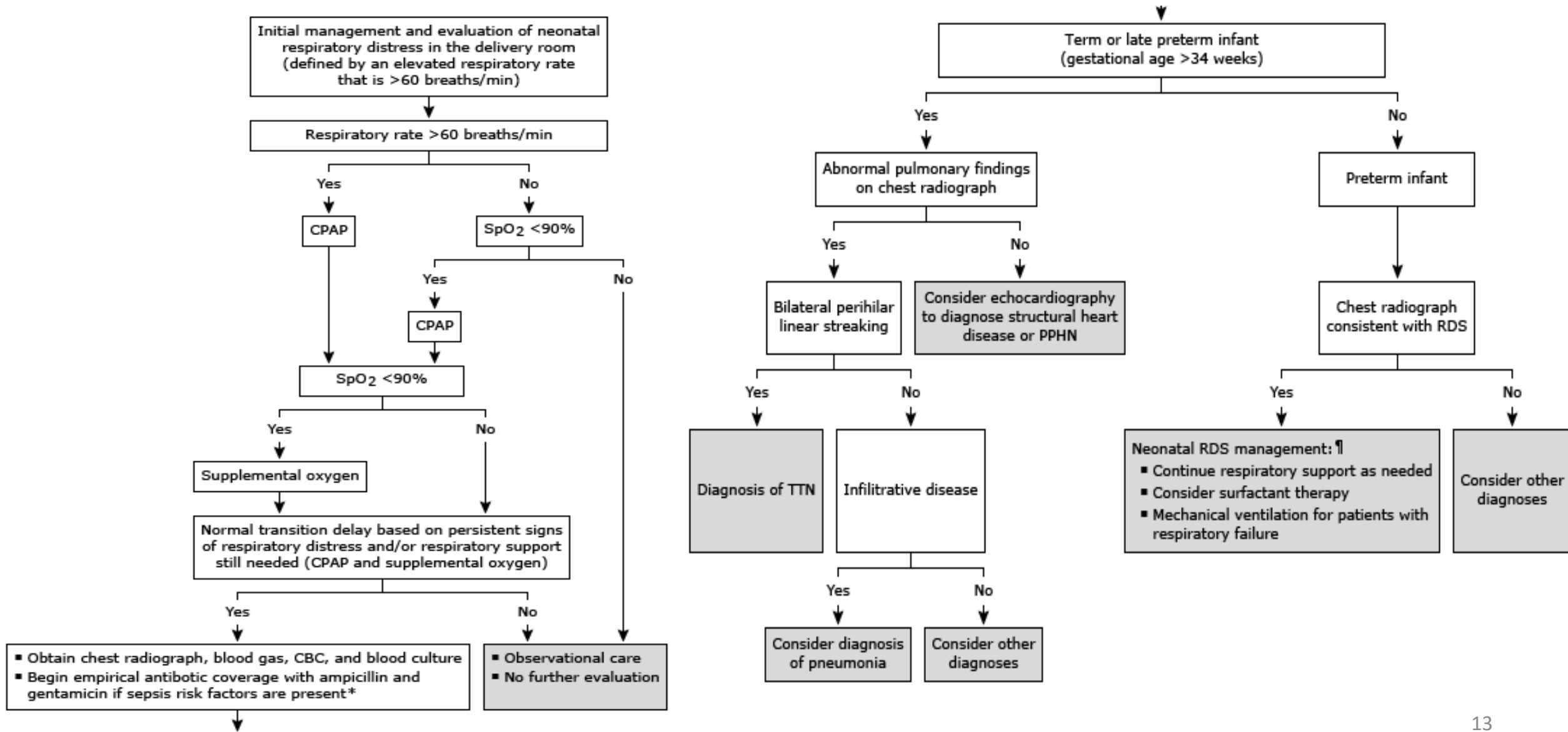
TABLE 5. **Differentiation of Cyanotic Heart Disease From Pulmonary Disease Among Infants in Respiratory Distress^a**

VARIABLE	CYANOTIC HEART DISEASE	PULMONARY DISEASE
History	Previous sibling with congenital heart disease Diagnosis of congenital heart disease by prenatal ultrasonography	Maternal fever MSAF Preterm delivery
Physical examination	Cyanosis Gallop rhythm or murmur Single second heart sound Large liver Mild respiratory distress	Cyanosis Severe retractions Split second heart sound Temperature instability
Chest radiograph	Increased heart size Decreased pulmonary vascularity (except in transposition of the great vessels or total anomalous pulmonary venous return)	Normal heart size Abnormal pulmonary parenchyma, such as total whiteout or patches of consolidation in pneumonia, fluid in the fissures in TTN or ground glass appearance in RDS
Arterial blood gas	Normal or decreased P_{aCO_2} Decreased P_{aO_2}	Increased P_{aCO_2} Decreased P_{aO_2}
Hyperoxia test	$P_{aO_2} < 150$ mm Hg	$P_{aO_2} > 150$ mm Hg (except in severe PPHN)
Echocardiography	Abnormal heart or vessels	Normal heart and vessels

MSAF=meconium-stained amniotic fluid; PPHN=persistent pulmonary hypertension of the newborn; RDS=respiratory distress syndrome; TTN=transient tachypnea of the newborn.

^aReproduced with permission from Aly et al. (23) Copyright 2014 by the American Academy of Pediatrics.

Approach ...Evaluation of neonatal respiratory distress



Specific causes of RD

Transient tachypnea of the newborn (TTN)

❖ Transient tachypnea of the newborn (TTN)

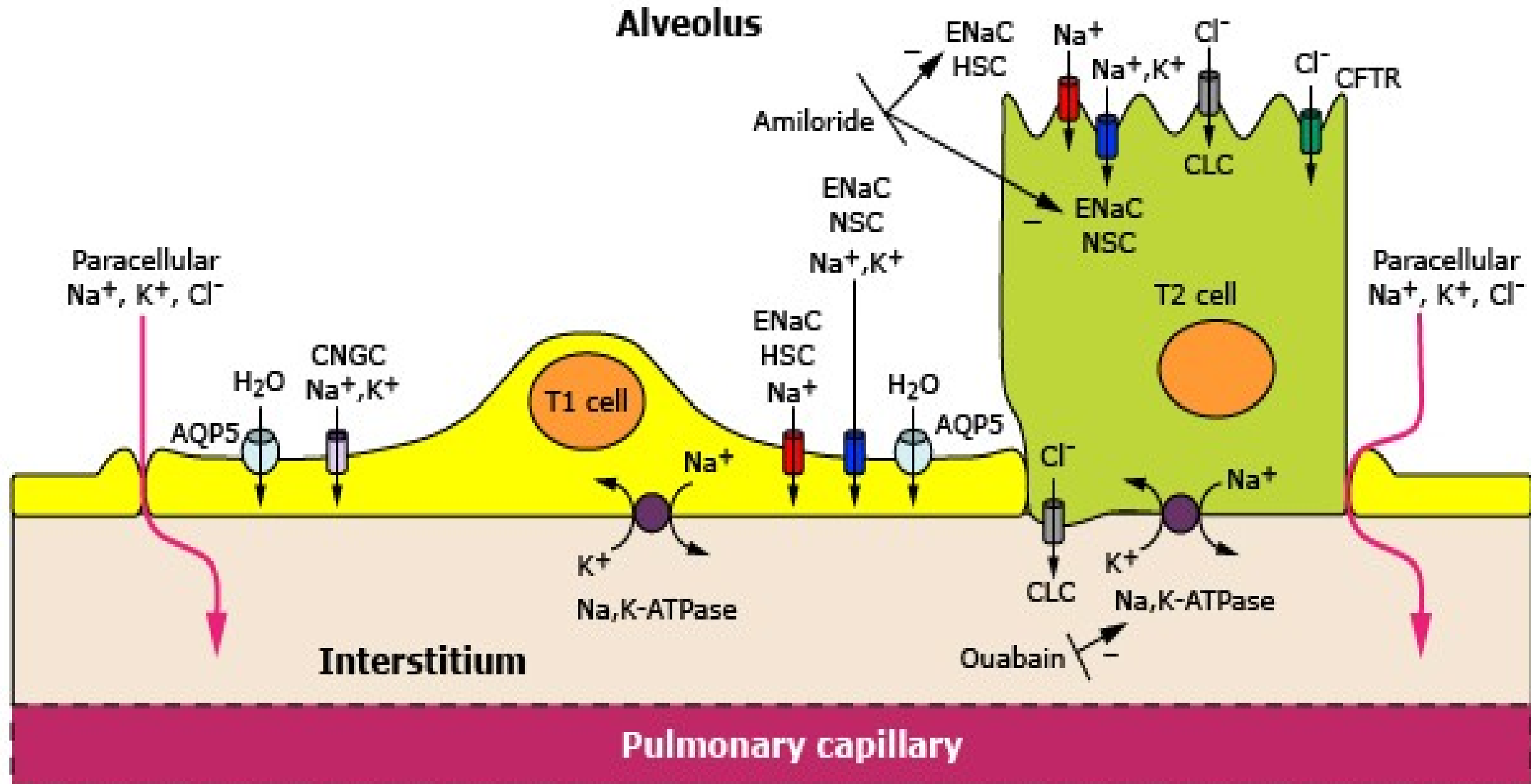
- Results from delayed clearance of fetal lung fluid
- Affects infants born at *late preterm or term gestation*.
- Is characterized by tachypnea with signs of **mild** respiratory distress including retractions and **cyanosis**; **decreased oxygen saturation** is usually alleviated by supplemental oxygen with $\text{FiO}_2 < 0.4$.

TTN.....

❖ II. PATHOPHYSIOLOGY.

- The lungs must switch from a **secretory** mode *in utero* to an **absorptive** mode.
- **Amloride-sensitive sodium channels** expressed in the apical membrane of the alveolar epithelium play an important role in lung fluid clearance.
- *Adrenergic stimulation* and other changes near birth lead to passive transport of sodium through the epithelial sodium channels
- **Thyroid hormone** also contributes to fetal lung fluid clearance by
 - increasing Na⁺/K⁺-ATPase activity and
 - β-adrenergic receptor stimulation in lungs and lymphatics.
- Interstitial lung fluid pools in perivascular cuffs of tissue and in the interlobar fissures and is then cleared into pulmonary capillaries and lung lymphatics.

PATHOPHYSIOLOGY



PATHOPHYSIOLOGY ...

- *Trans pulmonary hydrostatic pressure gradient* created by inspiratory effort is as important as the molecular mechanisms for clearance of airway fluid.
- Disruption or delay in clearance of fetal lung fluid results in the transient **pulmonary edema** that characterizes TTN.
- Compression of the compliant airways -- airway obstruction, air trapping, and ventilation–perfusion mismatch.
- Functional residual capacity may be reduced due to obstruction, whereas thoracic gas volume may increase secondary to air trapping.
- **Decreased surfactant function and reduced nitric oxide (NO) .**

TTN..

❖ III. EPIDEMIOLOGY.

- The incidence of TTN is 0.3% to 0.6% of term deliveries and 1% of preterm deliveries.
- **Risk factors -**
 - Cesarean delivery with or without labor, precipitous birth, and **preterm** birth
 - maternal hypotension during spinal anesthesia increase the risk of TTN.
 - Male gender and family history of asthma (especially mother).
 - **Genetic polymorphisms in β -adrenergic receptors** in alveolar type II cells
 - Macrosomia, maternal diabetes, pregnancy-induced hypertension, and multiple gestations
- Antenatal corticosteroids given to mothers at risk for late preterm births (34 to 36 weeks, 6 days) and prior to an elective cesarean section at ≥ 37 weeks' gestation reduce the risk of TTN.
- Onset of labor, oxytocin exposure, and **term** gestation provide some degree of **protection**.

TTN..

❖ IV. CLINICAL PRESENTATION.

- **Tachypnea** within the first 6 hours after birth
- Mild to moderate respiratory distress with retractions, **grunting**, nasal flaring, and/or **mild cyanosis** that usually responds to supplemental oxygen at <0.40 FiO₂.
- Increased **anteroposterior diameter** of the chest (**barrel-shaped**) due to hyperinflation, which may also push down the *liver and spleen, making them palpable*.
- **Auscultation** usually reveals **good** air entry, and crackles may or may not be appreciated.
- Signs of TTN usually persist for 12 to 24 hours in cases of mild disease but can last up to 72 hours in more severe cases.

TTN..

❖ V. DIFFERENTIAL DIAGNOSIS.

- Pneumonia/sepsis, RDS, pulmonary hypertension, Meconium aspiration
- Cyanotic congenital heart disease
- Congenital malformations (e.g., congenital diaphragmatic hernia, congenital pulmonary airway malformation)
- Central nervous system injury (subarachnoid hemorrhage, hypoxic-ischemic encephalopathy) causing central hyperventilation
- Pneumothorax
- Polycythemia, and metabolic acidosis.

TTN..

❖ VI. EVALUATION

A. History and physical examination.

- A careful history identifies elements such as prematurity, infectious risk factors, meconium-stained liquor.
- findings on physical examination such as cardiac or neurologic abnormalities may lead to a more targeted investigation.

TTN..

❖ B. Radiographic evaluation.

- Prominent **perihilar streaking (sunburst pattern)** due to **engorgement of periarterial lymphatics** that participate in the clearance of alveolar fluid.
- Coarse, fluffy densities may reflect alveolar edema.  fluid in the alveoli.
- **Hyperaeration** with widening of intercostal spaces, *mild cardiomegaly*, widened and fluid-filled interlobar fissure, and mild *pleural effusions* may also be observed.
- The radiographic findings in TTN usually **improve** by 12 to 18 hours and **resolve** by 48 to 72 hours.
- This rapid resolution helps **distinguish** the process from pneumonia and meconium aspiration.

Key TTN X-ray idea

Wet lungs + hyperinflation + perihilar streaking + fluid in fissures.



FIGURE 52-1 Transient tachypnea of the newborn. An anteroposterior radiograph of a term newborn on day 1 with transient tachypnea shows bilateral linear opacities (*arrow*) extending from the lung hila to peripheral lung fields.

TTN...

- ❖ **Lung ultrasound (LUS)** can differentiate TTN from RDS with **good specificity** but is not in common clinical use.
 - Absence of consolidation
 - Replacement of A-lines by B-lines
 - Numerous compact B-lines producing **“white lung,”** and **double-lung point (DLP)** sign are findings suggestive of TTN on LUS.
- Of note, the presence of **increased** pulmonary vascularity in the **absence** of cardiomegaly may represent total anomalous pulmonary venous return.

TTN...

❖ C. Laboratory evaluation.

- CBC and appropriate cultures can provide information concerning possible pneumonia or sepsis.
- An **arterial blood gas** may be used to determine the extent of hypoxemia and adequacy of ventilation.
- With persistent or severe hypoxemia, a cardiac evaluation should be considered.
- Respiratory alkalosis may reflect central hyperventilation due to CNS pathology or metabolic disorder.

TTN...

❖ VII. TREATMENT.

- Treatment is mainly supportive with provision of supplemental oxygen as needed.
- Invasive ventilation (intubation) is rarely required
- Early CPAP in the delivery room may reduce the severity and duration of respiratory distress in TTN.

❖ IX. PROGNOSIS.

- TTN is a self-limited process with no risk of recurrence, and the prognosis is excellent.
- Generally, there are no significant long-term residual effects.

Respiratory distress syndrome (RDS)

- Occurs primarily in premature infants;
 - Its incidence is inversely related to gestational age and birth weight.
 - It occurs in 60– 80% of infants <28 wk GA,
 - in 15–30% of those between 32 and 36 wk GA, and
 - rarely in those >37 wk GA.

RDS...

- The risk for development of **RDS increases** with;
 - Maternal diabetes, multiple births, cesarean delivery, precipitous delivery
 - Asphyxia, cold stress, and a maternal history of previously affected infants.
- The risk of RDS is **reduced** in pregnancies with
 - chronic or pregnancy-associated hypertension, maternal heroin use, prolonged rupture of membranes, and Antenatal corticosteroid prophylaxis.

RDS ...PATHOPHYSIOLOGY

❖ A. Lung maturity.

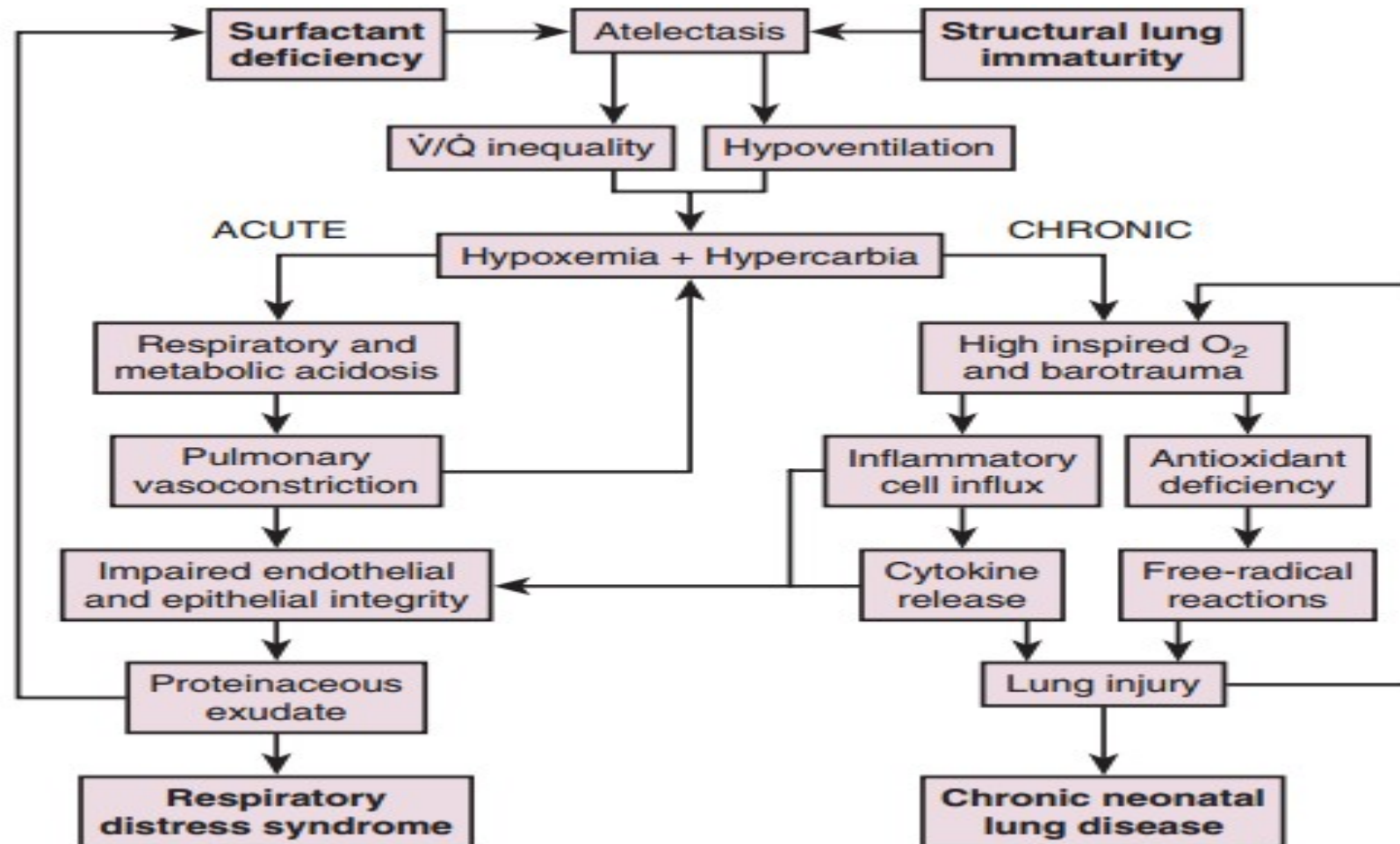
I. Factors that affect lung **maturation**

- **Maternal diabetes-** enhanced production of **fetal insulin** which **inhibits** the production of surfactant **phospholipids** important for surfactant **function**.
- **Labor-** Due to the production of endogenous maternal glucocorticoids, may **enhance lung** fluid clearance

RDS... PATHOPHYSIOLOGY

- **Mutations** - specifically **surfactant protein B** and **ABCA3**, result in severe RDS typically in **term** infants from either dysfunctional surfactant or severely limited production, respectively.
 - Infants with these mutations **die** without lung transplantation
- **Fetal sex.** **Male** infants are at a higher risk for RDS
 - due to the presence of circulating **weak fetal androgens** that inhibit the production of surfactant phospholipids.
- **Race.** Infants of African ancestry are at a **lower** risk for developing RDS,
 - due in part to the increased presence of protective **genetic** polymorphisms.

RDS... PATHOPHYSIOLOGY



RDS...pathophysiology

- ❖ Decreased surfactant production
- ❖ Presence of *hyaline membranes* located at *junction* of dilated respiratory bronchioles and dilated alveolar ducts
- ❖ Hyaline membranes contain *cellular debris from injured epithelium* and *fibrinous matrix components*
- ❖ Minimally aerated lungs, diffuse alveolar atelectasis
- ❖ Alveolar atelectasis, hyaline membrane *formation*, and *interstitial edema* make the lungs less compliant in RDS, so **greater pressure is required to expand the alveoli and small airways.**

❖ Surfactant deficiency

- high surface tension leading to instability of the lung at end-expiration, low lung volume, and decreased compliance.
 - hypoxemia due to a mismatch between ventilation and perfusion
- leads to lung **inflammation** and respiratory **epithelial injury**, which may result in **pulmonary edema and increased airway resistance**. These factors further exacerbate lung injury and worsen lung function. At the same time, abnormal fluid absorption results in inefficient clearing of liquid in the injured lung, leading to edema lung that also impedes gas exchange.

❖ Inflammation and lung injury —

- **surfactant deficiency** causes atelectasis that may lead to injury of the respiratory epithelium and the alveolar capillary endothelium, which can trigger a cytokine-mediated inflammatory response.
- **positive pressure ventilatory** support or excessive oxidant exposure .
- Lead to accumulation of protein-rich pulmonary fluid that can deactivate any surfactant that is present, thereby further exacerbating the underlying surfactant deficiency.

- ❖ Pulmonary edema — occurs because of the following contributory factors:
 - Inflammation and lung injury.
 - Reduced pulmonary fluid absorption – In preterm infants, an inadequate number of ENaC may result in fluid retention
 - Low urine output – contributing to fluid retention in the first few days, which may exacerbate pulmonary edema.
 - Infants recovering from RDS typically have a spontaneous diuresis on the second to fourth day, followed by improved pulmonary function.

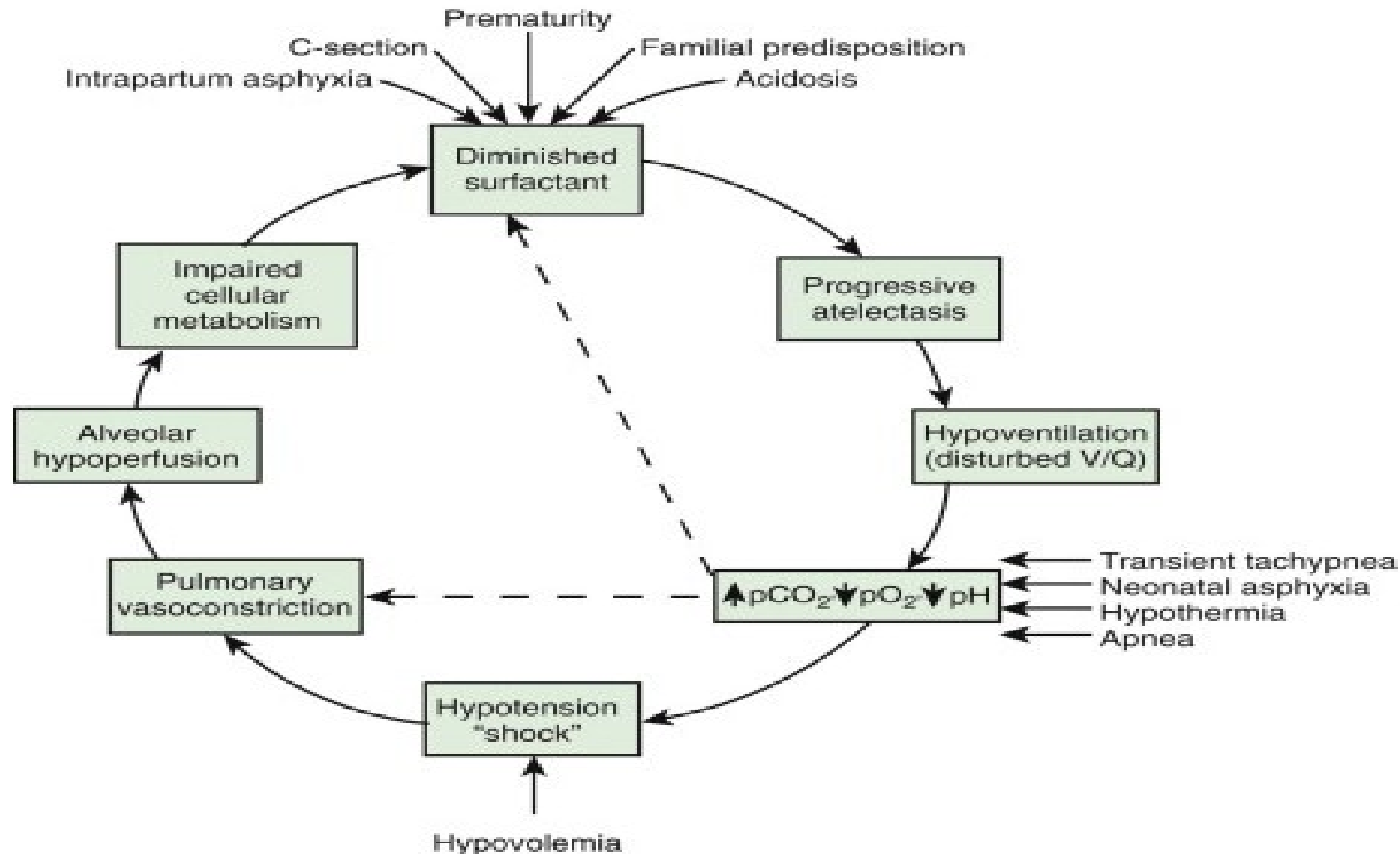
❖ Surfactant inactivation

- Meconium and blood in the alveoli
- Proteinaceous edema and inflammatory products

❖ Pulmonary function and gas exchange

- low compliance and low lung volume (functional residual capacity)

RDS...pathophysiology (contributing factors)



DIAGNOSIS

A. Antenatal testing

1. Invasive testing (amniocentesis) is not done nowadays
2. Lecithin/sphingomyelin (L/S) ratio and the presence of lamellar bodies in amniotic fluid were used in the past.

B. Clinical features.

- a preterm infant, typically <34 weeks of gestation, with signs of respiratory distress that develop soon after birth
- tachypnea, nasal flaring, subcostal and intercostal retractions, cyanosis, and an expiratory grunting

RDS...clinical manifestations

- Most infants present with SnSx **either in the delivery room or within the first 6 hours** after birth.
- The uncomplicated clinical course is cxzed by a **progressive worsening of symptoms with a peak severity by days 2 to 3** and **onset of recovery by 72 hours**
- Blood gas measurement will demonstrate hypoxemia and hypercarbia.
 - **Decreased** or absent air entry on auscultation

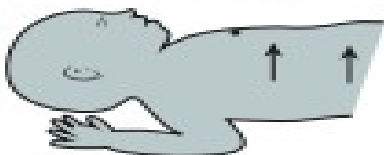
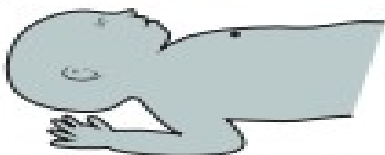
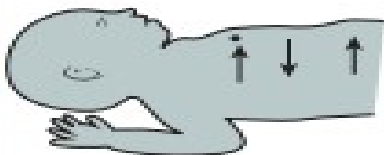
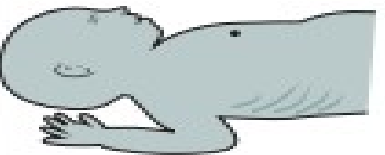
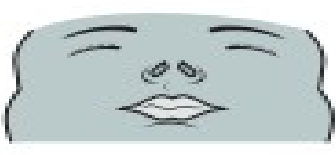
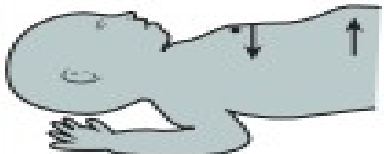
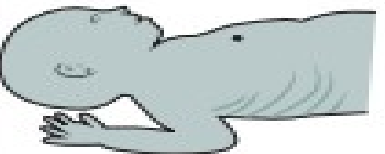
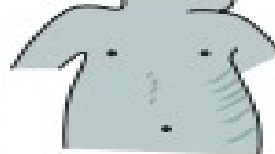
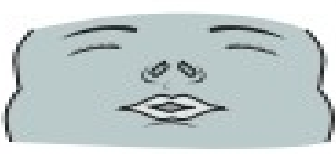
C/Manifestation ...Physiologic maneuvers to establish FRC

1. physiologic maneuvers to establish functional residual capacity (FRC) and optimize gas exchange -result in characteristic signs/symptoms of RDS.
- **Retractions.** To maximize the **negative inspiratory pressure** and thus lung inflation, affected infants use accessory muscles of breathing in addition to strong diaphragmatic contractions.
 - **Grunting** - is **active exhalation** against a partially closed glottis and results in a pressure gradient at the level of the vocal cords that provides expiratory distending pressure to stabilize patent but surfactant-poor alveoli.
 - **Hypoxia.** Need to give more oxygen (higher FiO₂) is the **simplest marker of severity of RDS** in preterm neonates with respiratory distress.
 - Most neonatologists recommend surfactant administration for preterm babies requiring 0.3 FiO₂ on optimal CPAP.

C/Manifestation ...Physiologic maneuvers to establish FRC

- **Tachypnea.** Inadequate FRC leads to inadequate tidal volumes. **To maintain minute ventilation** (the product of tidal volume and respiratory rate), infants with RDS increase the respiratory rate.
- **Flaring of the alae nasi.** Flaring of the alae nasi reduces the resistance to air flow through the upper airways.
- **Silverman–Andersen Score** - is a useful clinical severity grading score.
 - Higher the score (≥ 5), more is the work of breathing and higher the need for respiratory support to improve FRC.

Table 33.1. Silverman–Andersen Scoring to Assess the Severity of Respiratory Distress

	Upper chest	Lower chest	Xiphoid retracts	Nares dilate	Expiratory grunt
Grade 0	 Synchronized	 No Recessions	 None	 None	 None
Grade 1	 Lag on inspiration	 Just visible recessions	 Just visible	 Minimal	 Stethoscope only
Grade 2	 See-saw	 Marked Recessions	 Marked	 Marked	 Naked ear

Source: Silverman WA, Andersen DH. A controlled clinical trial of effects of water mist on obstructive respiratory signs, death rate and necropsy findings among premature infants. *Pediatrics* 1956;17:1–10.

Table :21:3 Downes' score for assessment of Respiratory Distress

Downes' score	0	1	2
Reparatory rate	<60Breaths per min	60-80	>80
Retractions	None	Mild	Severe
Cyanosis	None	Cyanosis relieved by oxygen	Cyanosis not relieved by oxygen
Grunting	None	Audible with stethoscope	Audible with ear
Air entry	Good bilaterally	Mildly decreased	Markedly decreased

Investigation X-RAY

C . Radiographic evidence.

- Typical radiographic findings result from homogeneous atelectasis of alveoli (**homogenous white lung**) with contrast “black” bronchi that do not collapse (**air bronchograms**: black bronchi stand out on the white background).
- Low lung volumes (rib spaces) are most evident before initiating CPAP/ ventilation.
- In severe RDS, the cardiac border and diaphragm border are not visible; the airless “white lung” merges with the “white” heart and liver shadows.

Investigation X-RAY ...

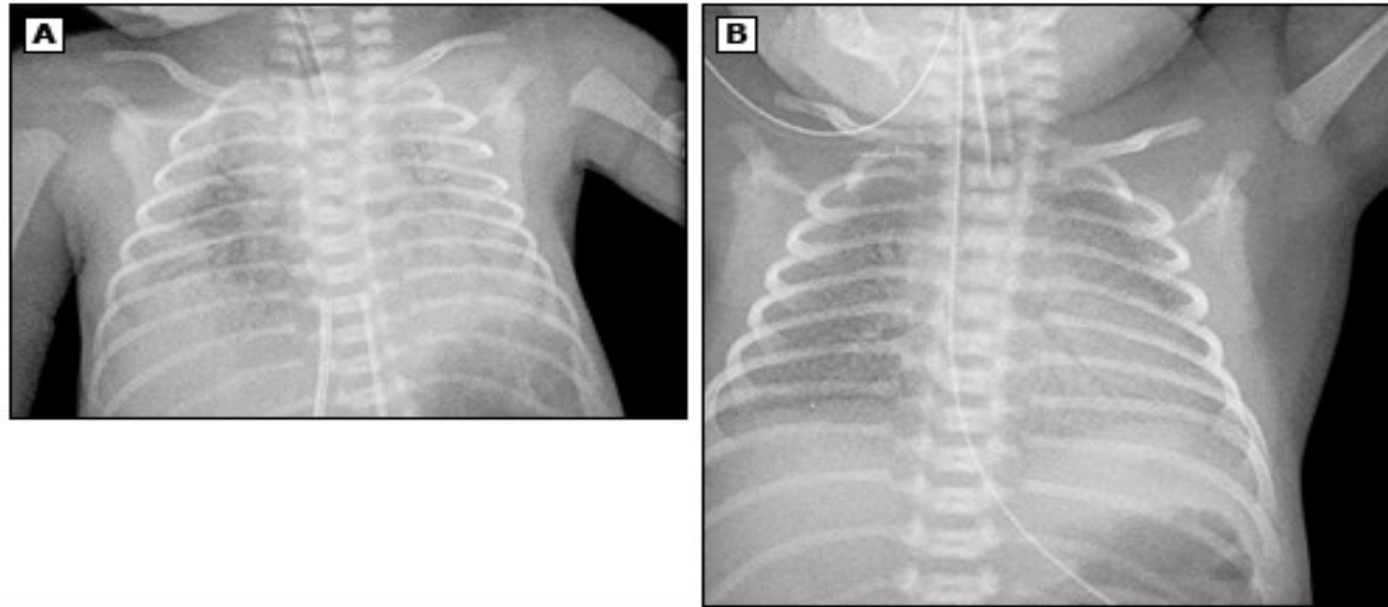
- **X-RAY PRESENTATION**

The four characteristic features of the x-ray presentation in RDS are:

1. Reduced lung volume
2. Air bronchograms
3. Reticulogranularity
4. Increased lung opacification

Investigation X-RAY ...

Chest radiograph of neonatal respiratory distress syndrome



Two radiographs that demonstrate severe (A) and moderate (B) neonatal respiratory distress syndrome. Both demonstrate the characteristic low lung volumes, and diffuse reticulogranular ground glass appearance with air bronchograms.

Investigation U/S

Lung ultrasound

Differentiate RDS from transient tachypnea of the newborn (TTN) and other neonatal respiratory disorders presenting at birth.

The main findings on the lung ultrasound are;

- **Consolidation** with air bronchogram
- Irregular and thickened **pleural line**
- Compact B-lines from the base to the apex
- A-line disappearance on **the transthoracic view** and
- Diffuse retro diaphragmatic hyper echogenicity replacing the normal diaphragm echo complex on the trans abdominal view

RDS...DDX

❖ DIFFERENTIAL DIAGNOSIS

- **A. Transient Tachypnea of Newborn (TTN)**
- **B. Pneumonia.**
- **C. Obstructed total anomalous pulmonary venous connection (TAPVC).**
 - ✓ a reticulogranular pattern to the lungs similar to RDS. **No response to surfactant, normal-volume lungs**, and 2D echo are useful in making a diagnosis
- **D. Genetic disorders of the surfactant system.**
 - Genetic mutations in surfactant protein **B and ABCA3** can result in an RDS picture in the immediate newborn period.
 - However, the infant shows short-lived **or no response** to the administration of artificial surfactant.
- **E. Disordered lung development.**
 - This category includes alveolar capillary dysplasia with malalignment of the pulmonary veins, congenital alveolar dysplasia, and brain–heart–lung disease due to mutations in NKX2.1/TTF1

RDS...

❖ PREVENTION

- **ANS** for pregnant women at 24 to 34 weeks of gestation with preterm labor (PTL) or planned preterm delivery.
- A **complete** course of ANS is considered to be
 - **Betamethasone** at 12 mg intramuscular (IM) **q24h** × 2 doses OR
 - **Dexamethasone** 6 mg IM q12h × 4 doses,
 - Course completed at least 24 hours before the delivery.
- Meta-analyses have not demonstrated superiority of one drug over the other.
- No contraindications exist to treatment.
- ANS must be given even if delivery is **imminent**.

RDS..mgt

❖ MANAGEMENT

- The key principles of treatment of RDS are to establish and maintain FRC, at the earliest.

➤ A. CPAP –

- ✓ Application of CPAP via binasal prongs, nasal mask, or face mask
- ✓ Early CPAP **reduces** the need to intubate and give surfactant and has potential in decreasing the incidence of bronchopulmonary dysplasia (BPD).

➤ B. Heated humidified high-flow nasal cannula (HHHFNC).

RDS...mgt

❖ C. Restore alveolar surfactant.

- 1. Prophylaxis versus treatment.
- 2. Surfactant preparations.
- 3. Dosing and dosing interval.
 - Surfactant is administered to achieve a phospholipid dose that is at least 100 mg/kg.
- 4. Administration. A surfactant is typically given to infants with RDS through an endotracheal tube (ETT)

RDS...mgt

❖ **D. Ensure appropriate ventilation.**

- Although poor **oxygenation** is the most evident feature of RDS, atelectasis also reduces CO₂ gas exchange resulting in hypercarbia.

➤ **1. Mechanical ventilation**

- Babies with poor respiratory effort, recurrent apnea, or severe RDS (high FiO₂ need, high CO₂ despite optimal CPAP and surfactant) should be identified early and supported with mechanical ventilation.

❖ VII. OUTCOMES OF RDS.

- In the presurfactant era, RDS in preterm infants typically resolved at 2 to 4 days of age, often preceded by **spontaneous diuresis**.
- With the widespread use of ANS, early use of CPAP, and exogenous surfactant, the time course of RDS has become much shorter, often just few minutes to hours.
- In some babies, association of preterm birth with chorioamnionitis may affect the time to resolution.
- RDS in infants born at ≥ 32 weeks of gestational age and without other complications typically resolves fully **with no long-term pulmonary sequelae**.
- Infants < 32 weeks of gestational age are at risk for **BPD**; the risk increases with decreasing gestational age

Meconium Aspiration

I. BACKGROUND

A. Cause.

- Acute or chronic hypoxia and/or infection - passage of meconium *in utero*.
- Gasping by the fetus or newly born infant - aspiration of amniotic fluid contaminated by meconium.
- Meconium aspiration before or during birth - obstruct airways, interfere with gas exchange, and cause severe respiratory distress

MAS...

❖ B. Incidence.

- MSAF complicates approximately 10% to 15% of deliveries.
- Most babies with MSAF are 37 weeks or older, and many meconium-stained infants are **postmature** and SGA.
- Approximately 3% to 4% of neonates born through MSAF develop MAS in **Western** countries, but a higher incidence of around 11% to 28% is reported in the developing world.
- Approximately 30% to 50% of these infants require CPAP or mechanical ventilation.

MAS...PATHOPHYSIOLOGY

- **II. PATHOPHYSIOLOGY.**

- Meconium is a **sterile**, **thick**, **black-green**, **odorless** material that results from the accumulation of **debris** in the **fetal intestine** starting in the **third month of gestation**.
- The components of meconium includes;
 - **water (72% to 80%)**,
 - desquamated cells from the intestine and skin, gastrointestinal mucin, lanugo hair, fatty material from the vernix caseosa, amniotic fluid, intestinal secretions, blood group–specific glycoproteins, bile, and
 - enzymes including phospholipase A2.

MAS...PATHOPHYSIOLOGY

❖ A. Passage of meconium *in utero*.

- MSAF occurs more commonly in term or postterm pregnancies and rarely prior to 34 weeks' gestation.
- It may result from a post-term fetus with rising motilin levels and normal gastrointestinal function, vagal stimulation produced by cord or head compression, or *in utero* fetal stress.
 - The amniotic fluid that is thinly stained is described as *watery*.
 - The moderately stained fluid is opaque without particles, and
 - fluid with thick meconium with particles is called *pea soup*.
- The severity of MAS appears not to be related to thickness of MSAF.

MAS...PATHOPHYSIOLOGY

- **B. Aspiration of meconium.**
 - In the presence of fetal **stress**, gasping by the fetus can result in aspiration of meconium before, during, or immediately following delivery.
 - **Severe** MAS appears to be caused by pathologic intrauterine processes, primarily chronic *hypoxia*, *acidosis*, and *infection*.

MAS...PATHOPHYSIOLOGY

❖ C. Effects of meconium aspiration (C/F).

- Meconium staining nails, skin, umbilical cord
- Mechanically obstructs the small airways, causes
 - Increased AP diameter of the chest
 - atelectasis, and a “**ball valve**” effect with resultant air trapping and
 - **possible air leak.**
- leads to **vasospasm**, hypertrophy of the pulmonary arterial musculature, and pulmonary hypertension.
- One-third of infants with MAS develop **PPHN**, which contributes to the **mortality**
- Inhibits surfactant **function- RDS.**

MAS...PATHOPHYSIOLOGY

❖ D. Severity.

- Mild in infants requiring <40% oxygen for <48 hours and
- Moderate in infants requiring >40% oxygen for >48 hours without air leak.
- Severe in infants who require assisted ventilation and/or CPAP and is often associated with PPHN.

Table :21:3 Downes' score for assessment of Respiratory Distress

Downes' score	0	1	2
Respiratory rate	<60 Breaths per min	60-80	>80
Retractions	None	Mild	Severe
Cyanosis	None	Cyanosis relieved by oxygen	Cyanosis not relieved by oxygen
Grunting	None	Audible with stethoscope	Audible with ear
Air entry	Good bilaterally	Mildly decreased	Markedly decreased

MAS...PATHOPHYSIOLOGY

❖ E. Sequelae.

- in **term** infants- an increased risk of perinatal and neonatal **mortality**, severe acidemia, need for cesarean delivery, and **adverse neurologic outcome**.
- In **Preterm** infants - **similar** adverse effects as well as an increased incidence of severe intraventricular hemorrhage, cystic periventricular leukomalacia, and cerebral palsy.

MAS...PATHOPHYSIOLOGY

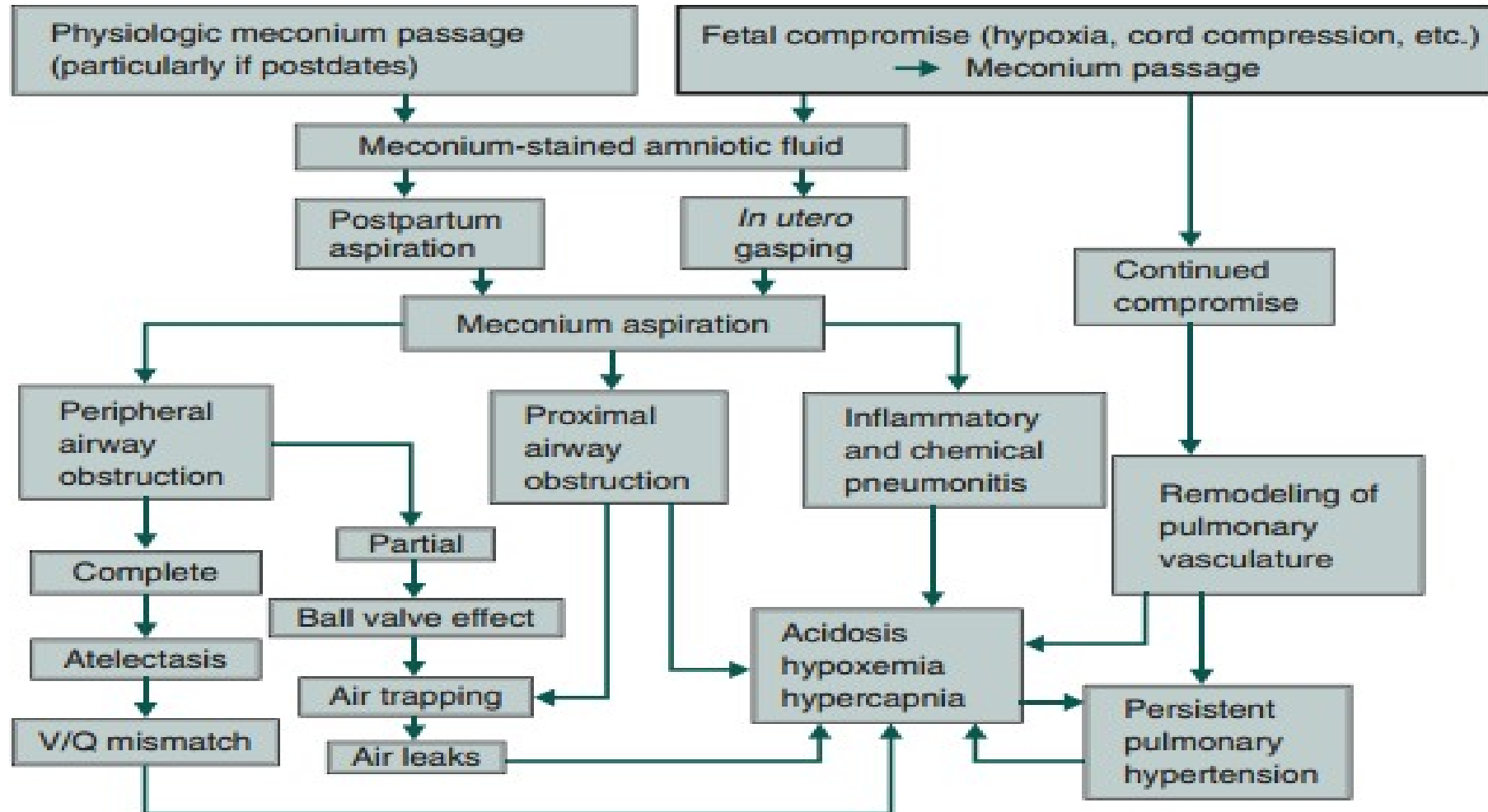


Figure 35.1. Pathophysiology of meconium aspiration. V/Q, ventilation–perfusion ratio. (From Wiswell T, Bent RC. Meconium staining and the meconium aspiration syndrome: unresolved issues. *Pediatr Clin North Am* 1993;40:955–981. Used with permission.)

MAS...Prevention

- III. PREVENTION OF MECONIUM ASPIRATION SYNDROME

- A. Prevention of the passage of meconium *in utero* (*risk factors*).

- Preeclampsia or increased blood pressure,
 - Chronic respiratory or cardiovascular disease, hypertension
 - heavy smokers.
 - Prolonged labor
 - Post maturity
 - diabetes,
 - Umbilical cord complications •
 - Intrauterine growth retardation (IUGR).

MAS...Prevention

- B. Amnioinfusion.
 - is associated with substantive improvements in the perinatal outcome **only** in settings where facilities for perinatal surveillance are **limited**.
 - It is not clear whether the benefits are due to dilution of meconium or relief of oligohydramnios.
 - In settings with **standard** peripartum surveillance, either amnioinfusion is ineffective or its effects are masked by other strategies to optimize neonatal outcome.
- C. Timing and mode of delivery.
 - **induction** as early as 41 weeks may help prevent MAS by avoiding the passage of meconium.
 - The **delivery mode** does not appear to impact the risk of aspiration

MAS... mgt

- IV. MANAGEMENT OF INFANTS DELIVERED THROUGH MECONIUM-STAINED FLUID.
 - **vigorous** - Oropharyngeal and nasopharyngeal suctioning on the perineum and routine tracheal intubation and aspiration of meconium are **not effective in preventing MAS**.
 - Non vigorous - intubation and endotracheal suctioning for meconium previously recommended, **currently is no longer recommended.**
 - -intubation and suction if the airway is obstructed
 - Trachial suctioning using a suction catheter inserted through the endotracheal tube
 - The recommended suction pressure should be less than 80 to 100 mm Hg.

MAS... mgt

❖ MANAGEMENT OF MECONIUM ASPIRATION SYNDROME

A. Observation.

- Infants born through MSAF are at risk for meconium aspiration pneumonia and should be observed closely for respiratory distress.
- 1. A **chest radiograph** may help determine those infants who are *most likely to develop respiratory distress*,
 - The classic roentgenographic findings are diffuse, asymmetric patchy infiltrates, areas of consolidation, often worse on the right, and hyperinflation.
- 2. Monitoring of **oxygen saturation** during this period aids assessment of the **severity** of the infant's condition and avoids hypoxemia.

MAS... mgt

- **B. Care for neonate with meconium aspiration syndrome**

- 1. The infant should be maintained in a **neutral thermal environment**, and tactile stimulation should be **minimized**.
- 2. **Blood glucose** and **calcium levels** should be assessed and corrected if necessary.
Severely depressed infants may have severe metabolic acidosis that may need to be corrected.
- 3. Infants may also require specific therapy for **hypotension** and **poor cardiac output**, including cardiotonic medications such as dopamine.

MAS... mgt

- 4. **Circulatory support with normal** saline or packed red blood cells should be provided in patients with **marginal oxygenation**.
 - In infants with substantial oxygen and ventilator requirements, we usually maintain a hemoglobin concentration above **15 g (hematocrit above 40%)**.
- 5. Renal function should be continuously monitored
- 6. We **avoid** chest physiotherapy because of the potential adverse effect of exacerbating PPHN.
- 7. Airway and oral **suctioning** may be required to facilitate airway clearance, but potential benefits must be balanced against the risk of hypoxic episodes and subsequent worsening of PPHN.

MAS... mgt

- **C. Oxygen therapy.**
- It is crucial to provide sufficient oxygen because repeated hypoxic insults may result in ongoing pulmonary vasoconstriction and contribute to the development of PPHN.

- **D. Assisted ventilation**

- 1. CPAP.**

- If FiO requirements exceed 0.40, a trial of CPAP may be considered.

- CPAP may sometimes aggravate air trapping and should be instituted with caution if *hyperinflation* is apparent clinically or radiographically.

MAS... mgt

- **2. Mechanical ventilation.**
 - Mechanical ventilation is indicated for excessive carbon dioxide retention ($\text{PaCO}_2 > 60 \text{ mm Hg}$) or for persistent hypoxemia ($\text{PaO}_2 < 50 \text{ mm Hg}$).
- **3. Inhaled nitric oxide (iNO) and extracorporeal membrane oxygenation (ECMO)** may be required for infants with refractory respiratory failure.

MAS... mgt

- **E. Medications**

- 1. **Antibiotics.**

- Although few infants with MAS have documented infections, the use of broad-spectrum antibiotics (e.g., ampicillin and gentamicin) is usually indicated in infants when an **infiltrate** is seen on chest radiograph.
- 2. **Surfactant.**
 - Endogenous surfactant **activity** may be inhibited by meconium and is a secondary cause of surfactant deficiency.
 - **Surfactant replacement** in MAS improves oxygenation, reduces the need for ECMO, and is recommended by the Committee on Fetus and Newborn of the American Academy of Pediatrics.
 - 3. **Corticosteroids.** - not routinely recommend

MAS... mgt

❖ F. Complications

- **1. Air leak.**
 - Pneumothorax or pneumomediastinum occurs in approximately 15% to 33% of patients with MAS
- **2. PPHN –**
 - is associated with MAS in approximately one-third of cases and contributes to the **mortality** associated with this syndrome.
 - In severely ill infants with MAS and PPHN, iNO reduces the need for ECMO.
- **3. Pulmonary sequelae.**
 - Approximately 5% of survivors require supplemental oxygen at 1 month and a substantial proportion may have an abnormal pulmonary function, including increased functional residual capacity, airway reactivity, and a higher incidence of pneumonia.



Thank you